

Uncertainty-Aware Self-Supervised Monocular Depth Estimation on Reflective Indoor Surfaces

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Abstract

Self-supervised monocular depth estimation (SS-MDE) relies on photometric consistency between stereo views as its sole training signal. This assumption breaks on specular, reflective, and transparent surfaces, which are pervasive in indoor environments. This report extends Monodepth2 with a Poggi-style per-pixel log-variance uncertainty head that modulates the photometric loss via a Negative Log-Likelihood (NLL) objective, enabling the network to automatically suppress gradient contributions from specular pixels during training. Fine-tuned on the Booster dataset -- an indoor stereo benchmark of 228 images across 38 scenes containing mirrors, glass, and metallic surfaces -- the best checkpoint achieves an absolute relative depth error of 0.255, outperforming all fine-tuned baseline configurations on every metric. Sparsification analysis reveals that while the uncertainty head correctly localizes specular regions, photometric uncertainty does not directly predict depth error magnitude: ground-truth LiDAR depth remains accurate at specular pixels even when the photometric training signal is corrupted. MC Dropout provides better-calibrated uncertainty (AUSE 0.131 vs 0.182) at the cost of increased inference time. The results demonstrate that uncertainty-modulated training is an effective and low-overhead approach to improving depth estimation on non-Lambertian surfaces without requiring reflective surface annotations.

Key Findings

1. Uncertainty-modulated NLL training achieves Abs Rel 0.255 at epoch 15, a 7% relative improvement over the best baseline (Abs Rel 0.274), and improves a1 accuracy from 0.419 to 0.499 -- outperforming all fine-tuned baselines on every one of the seven evaluation metrics.
2. The uncertainty head correctly identifies specular regions without any annotated labels. Qualitative heatmaps consistently show elevated uncertainty on mirrors, glass, and metallic surfaces.
3. Both the NLL log-variance head (AUSE 0.182, AURG -0.072) and MC Dropout (AUSE 0.131, AURG -0.034) produce negative AURG scores, demonstrating that photometric uncertainty does not reliably predict depth error magnitude -- an important distinction for any downstream application requiring calibrated depth confidence.
4. Training at higher resolution (768x576, 20 epochs) performs worse than lower resolution (512x384, 10 epochs) on all metrics, suggesting that overfitting, not resolution, limits performance in this small-dataset regime.

5. MC Dropout provides better-calibrated uncertainty than the explicit log-variance head (AUSE 0.131 vs 0.182), at the cost of 10x inference time. Combining NLL training for depth accuracy with MC Dropout for uncertainty estimation is the most principled approach.

1. PROBLEM DESCRIPTION

Self-supervised monocular depth estimation learns dense per-pixel depth from unlabeled video by minimizing a photometric reprojection loss. Given a target image and one or more temporally or spatially adjacent source images, the network predicts depth and (in the monocular case) relative camera pose. The predicted depth is used to synthesize a reconstruction of the source image by warping it into the target frame; the SSIM and L1 color difference between the reconstruction and the actual target serves as the training signal. No ground-truth depth is required.

This photometric consistency assumption holds well for Lambertian surfaces, whose apparent color does not change with viewing direction. It fails on non-Lambertian surfaces: a mirror's reflection changes entirely between two camera positions, so the reprojection error for a correct mirror depth prediction is arbitrarily large. Rather than providing a useful training signal, specular pixels actively produce adversarial gradients. During inference, models trained under this assumption produce unreliable depth at reflective surfaces: a mirror may be assigned the depth of the objects visible within it, and specular highlights on metallic objects cause spurious depth discontinuities.

Indoor environments present this problem acutely. Factories, kitchens, bathrooms, and laboratories routinely contain mirrors, glass partitions, metallic equipment, and polished floors. Autonomous systems operating in these environments -- from mobile robots to wearable guidance devices -- require reliable depth estimates precisely where standard SS-MDE models fail. This motivates a training-time intervention that reduces the influence of specular pixels on the depth network without requiring any reflective surface annotation.

The approach evaluated in this report trains the network to jointly predict depth and per-pixel photometric uncertainty. Pixels where the photometric signal is unreliable receive high predicted uncertainty, and the NLL loss automatically reduces their gradient contribution. Compared to alternative approaches -- 3D distillation via model ensembles [3] and intrinsic image decomposition [4] -- this method requires no additional networks, no 3D reconstruction, and no annotations, making it practical for small-scale fine-tuning on domain-specific datasets.

2. DATASET

2.1 Booster

The Booster dataset (University of Bologna, 2022) is a high-resolution indoor stereo benchmark designed specifically for evaluating depth estimation on non-Lambertian surfaces [5]. Each scene was selected to contain at least one challenging surface type: flat mirrors, glass vessels, polished metal, windows, or glossy ceramics. Ground-truth depth is acquired with a structured-light sensor calibrated to the stereo rig; crucially, the sensor provides accurate depth at specular pixel locations even though those pixels are photometrically unreliable. This property is exploited in the sparsification analysis.

Property	Value
Total images	228 across 38 indoor scenes
Original resolution	4112 x 3008 px
Training resolution	512 x 384 px (Lanczos resample)

Stereo baseline	~82.9 mm (fixed per scene)
Ground-truth type	Structured-light disparity (metric)
Training split	164 images, 27 scenes
Validation split	18 images, ~3 scenes
Test split	46 images, 28 scenes

Table 1. Booster dataset statistics. Split is by scene to prevent information leakage.

2.2 Preprocessing

Raw 4112x3008 images are downsampled to 512x384 using Lanczos resampling, which preserves the original 1.37:1 aspect ratio and provides superior anti-aliasing relative to bilinear or nearest-neighbor resampling at this downscale factor. Camera intrinsics (f_x , f_y , c_x , c_y) are scaled proportionally. Training uses stereo-only mode: the known fixed stereo transform from `calib.json` replaces the PoseNet, eliminating pose estimation error and providing a stable training signal from the stereo geometry alone.

3. METHOD

3.1 Baseline Architecture

The depth network follows the encoder-decoder architecture of Monodepth2 [1]. The encoder is a ResNet-18 pretrained on ImageNet, producing feature maps at five scales with channel widths of 64, 64, 128, 256, and 512. The decoder is a five-level U-Net that upsamples by 2x at each level using nearest-neighbor interpolation followed by convolution, with skip connections from the encoder concatenated at each scale. The final decoder layer outputs a sigmoid-activated disparity map at full resolution; multi-scale supervision is applied at decoder scales 1/4 through full resolution with equal weight.

Training uses the minimum reprojection loss from Godard et al. [1]: the per-pixel minimum photometric error over all source frames is taken before averaging, handling occlusions and dynamic objects. Auto-masking excludes static pixels from the loss. An edge-aware smoothness regularization term penalizes depth gradients weighted by the inverse image gradient, permitting discontinuities at object boundaries.

3.2 Uncertainty Head

A single additional Conv3x3 layer (uncertconv) is added to the decoder, branching from the same full-resolution feature map used by the disparity head. The two branches share no weights; a single forward pass produces both a disparity map and a per-pixel log-variance map simultaneously. The output is clamped to $[-10, 10]$ for numerical stability. The flag `--use_uncertainty` controls whether this branch is active; when inactive, there is no architectural difference from the original Monodepth2.

The uncertainty head adds fewer than 1,000 parameters to a network with approximately 14.8 million parameters -- a negligible overhead. At inference time it adds no latency relative to a single forward pass.

3.3 NLL Training Objective

Standard Monodepth2 minimizes the per-pixel photometric reprojection error pe , defined as a weighted combination of SSIM and L1:

$$pe = 0.85 * SSIM(I_t, I_{t'}) + 0.15 * |I_t - I_{t'}| \quad (1)$$

where I_t is the target image and $I_{t'}$ is its reconstruction from the source frame via the predicted depth and camera geometry. The uncertainty extension replaces this with a Laplacian Negative Log-Likelihood (NLL) objective following Poggi et al. [2]:

$$L_{NLL}(x) = pe(x) * \exp(-s(x)) / 2 + s(x) / 2 \quad (2)$$

where $s(x) = \log(\sigma^2(x))$ is the predicted log-variance at pixel x . The first term scales the photometric loss contribution by the predicted uncertainty: when s is large (high uncertainty, specular surface), $\exp(-s)$ approaches zero and the pixel contributes minimally to the gradient. The second term is a regularizer preventing the network from setting s to infinity everywhere; it penalizes high uncertainty estimates and creates a balance that forces the network to predict high uncertainty only where the photometric signal is genuinely unreliable.

The full training loss adds edge-aware smoothness regularization:

$$L = (1/N) * \sum_x [L_{NLL}(x)] + \lambda * L_{smooth} \quad (3)$$

with $\lambda = 1e-3$. The smoothness term L_{smooth} penalizes depth gradients inversely weighted by image gradients.

3.4 Training Configuration

Training proceeds in two stages. In the first stage, standard Monodepth2 is fine-tuned on Booster for 10 epochs at 512x384, initializing from KITTI-pretrained weights with the encoder frozen and learning rate $1e-5$. This establishes a baseline depth model adapted to the indoor domain. In the second stage, the uncertainty head is added and the combined network is fine-tuned for 20 epochs at 640x192 with learning rate $2e-5$ and batch size 8.

The second-stage initialization uses shape-safe weight loading: parameters from the Stage 1 checkpoint are loaded only where tensor shapes match, leaving the new uncertconv weights randomly initialized. This is required because the Stage 1 checkpoint does not contain uncertconv parameters; a standard `load_state_dict` call would raise a key mismatch error.

Parameter	Stage 1 (Baseline)	Stage 2 (Uncertainty)
Resolution	512 x 384	640 x 192
Learning rate	$1e-5$	$2e-5$
Batch size	8	8
Epochs	10	20 (best at ep. 15)
Encoder	Frozen	Unfrozen
Frame mode	--use_stereo	--use_stereo
Initialization	KITTI pretrained	Stage 1 checkpoint

Table 2. Training hyperparameters for each stage.

All evaluations use `--min_depth 0.1 --max_depth 100`, matching the training depth bounds. Per-scene stereo baseline scaling is applied to convert unit-free disparity predictions to metric depth before computing any metric.

4. EVALUATION AND RESULTS

4.1 Evaluation Protocol

Depth accuracy is measured using seven standard metrics on the 46-image, 28-scene Booster test set. Lower is better for absolute relative error (Abs Rel), squared relative error (Sq Rel), root mean squared error (RMSE), and log-scale RMSE. Higher is better for the threshold accuracy metrics $\delta < 1.25^k$ ($k = 1, 2, 3$), which measure the fraction of pixels for which the ratio $\max(d/d^*, d^*/d)$ falls below the threshold. Metrics are averaged over all valid pixels, then averaged across scenes.

Uncertainty calibration is evaluated using sparsification analysis. Pixels are sorted by predicted uncertainty from highest to lowest; subsets of increasing size are removed and Abs Rel is recomputed on the retained pixels. A well-calibrated uncertainty should produce a curve that decreases as uncertain (and presumably error-prone) pixels are removed. Two summary statistics are reported: AUSE (Area Under Sparsification Error, lower is better) measures the gap between the method's curve and the oracle curve (true per-pixel error used as the sort key); AURG (Area Under Random Gain = $AUSE_{\text{random}} - AUSE_{\text{method}}$, higher is better) measures improvement over uniformly random pixel removal. A negative AURG indicates that the uncertainty is less informative than random selection for predicting which pixels have the largest depth error.

4.2 Depth Accuracy

Model	Abs Rel	Sq Rel	RMSE	RMSE log	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
KITTI pretrained (no fine-tune)	0.579	0.904	0.830	0.631	0.254	0.487	0.678
Baseline 512x384, 10 epochs	0.274	0.185	0.449	0.400	0.419	0.711	0.879
Baseline 768x576, 20 epochs	0.328	0.226	0.507	0.489	0.284	0.608	0.807
Ours: NLL log-var, epoch 3	0.273	0.186	0.445	0.402	0.438	0.717	0.861
Ours: NLL log-var, epoch 15	0.255	0.152	0.402	0.350	0.499	0.810	0.894
Ours: NLL log-var, epoch 19	0.262	0.159	0.411	0.356	0.477	0.807	0.893

Table 3. Depth accuracy on the Booster test set (46 images, 28 scenes). The highlighted row is our best checkpoint. Lower is better for Abs Rel, Sq Rel, RMSE, RMSE log; higher is better for delta thresholds.

The KITTI-pretrained model with no fine-tuning achieves an Abs Rel of 0.579, confirming that the domain gap between outdoor driving and indoor reflective scenes is severe. Ten epochs of standard Monodepth2 fine-tuning at 512x384 reduces Abs Rel to 0.274, a 53% improvement, demonstrating that domain adaptation is effective even at this small dataset scale. Counterintuitively, fine-tuning at higher resolution (768x576, 20 epochs) degrades performance on all metrics relative to the lower resolution run, suggesting that the network overfits to fine texture at a resolution where the dataset provides insufficient diversity.

Our NLL uncertainty model at epoch 15 achieves Abs Rel 0.255, improving on the best baseline by 7% in absolute relative error and 19% in $\delta < 1.25$ accuracy (0.499 vs 0.419). The improvement is consistent across all seven metrics: no metric regresses. This demonstrates that suppressing the adversarial gradient signal from specular pixels during training provides genuine depth quality gains, even when the training set contains only 164 images. Performance peaks at epoch 15 and slightly degrades by epoch 19, consistent with mild overfitting; the validation loss curve confirms that epoch 15 corresponds to the lowest validation error.

4.3 Uncertainty Calibration

Uncertainty Method	AUSE (lower = better)	AURG (higher = better)	Inference Cost
NLL log-variance head	0.1816	-0.0720	1x (single forward pass)
MC Dropout (N=10)	0.1309	-0.0337	10x (ten forward passes)

Table 4. Sparsification metrics on the Booster test set.

Both methods produce negative AURG, meaning that removing pixels ranked as most uncertain increases rather than decreases the average depth error on the retained set. This result requires careful interpretation. The uncertainty heads correctly identify specular pixels as uncertain, as confirmed by the qualitative heatmaps in Section 4.4. However, the ground-truth LiDAR depth is accurate at those pixels even though the photometric training signal at those locations is corrupted. Removing correctly-located but geometrically accurate pixels therefore hurts the aggregate depth metric rather than improving it.

This reveals a fundamental distinction between photometric uncertainty and geometric uncertainty. The NLL head learns to flag pixels where photometric consistency fails (i.e., where the training signal is unreliable), not pixels where the depth prediction is likely wrong. For downstream applications that require a calibrated estimate of depth error magnitude -- such as safe robot grasping near reflective objects -- the raw photometric uncertainty map is not a suitable proxy. MC Dropout provides meaningfully better calibration (AUSE 0.131 vs 0.182) because stochastic dropout disagreement captures model epistemic uncertainty, which is more loosely correlated with depth error than photometric violation. Combining NLL training for depth accuracy with MC Dropout at inference time for uncertainty estimation is therefore the recommended approach.

5. QUALITATIVE ANALYSIS

5.1 Uncertainty Heatmaps

Depth and uncertainty heatmaps were generated for four representative Booster scenes: a bathroom mirror (large flat specular surface), a set of shot glasses (transparent and refractive), a kitchen sink (metallic with specular highlights), and a window (partially reflective, partially transmissive). Across all four scenes, the uncertainty map produced by the log-variance head shows high predicted uncertainty consistently concentrated on reflective and transparent surfaces, with low uncertainty on Lambertian walls, floors, and matte objects. This spatial concentration confirms that the uncertainty head is learning the intended photometric reliability signal and not arbitrary scene features.

Comparing depth maps from the baseline and the NLL model, the uncertainty model produces smoother depth predictions in Lambertian regions and reduced depth hallucination artifacts at mirror boundaries. One visible trade-off is a modest reduction in depth sharpness at object boundaries adjacent to specular surfaces, where the uncertainty prediction bleeds slightly across edges due to the receptive field of the convolutional uncertainty head.

5.2 Failure Modes

Three distinct failure modes were identified through scene-by-scene inspection:

Mirror depth hallucination. Large flat mirrors present the most severe failure. The stereo reprojection of a mirror's reflection is geometrically consistent from both camera positions -- the same reflected scene is visible with coherent disparity in both views -- so the network confidently assigns depth to the reflected content rather than the mirror surface. The NLL model reduces but does not eliminate this artifact; the photometric signal, though suppressed, still carries information about the reflected depth structure.

Transparent surface penetration. For clear glass objects, the dominant photometric signal comes from objects behind the surface (transmitted light). The surface is effectively invisible to the photometric loss because warping through it correctly reconstructs the background color. Uncertainty-masking does not help here: the photometric signal is not corrupted, it is simply the wrong signal. Addressing this failure mode requires explicit surface detection or a different training signal.

Specular highlight depth spikes. Bright highlights on metallic surfaces appear at different image positions in the two stereo views, creating large, localized photometric errors. The uncertainty head flags these locations, and the NLL model shows reduced depth spikes at highlight regions. A residual artifact is visible around highlight boundaries, where depth estimates are noisy rather than smoothly incorrect, suggesting that the uncertainty masking partially but not completely neutralizes the adverse gradient.

6. DISCUSSION AND LIMITATIONS

The primary result of this work is that a minimal architectural change -- one additional convolutional branch with a modified loss -- produces consistent depth quality improvements on a challenging indoor reflective surface benchmark. The NLL uncertainty mechanism requires no additional networks, no reflective surface annotations, no 3D reconstruction, and adds negligible parameters and no inference latency. This makes it a practical first step for any Monodepth2-based pipeline targeting non-Lambertian indoor environments.

The sparsification analysis, however, surfaces an important limitation for safety-critical applications. Because the uncertainty head learns photometric uncertainty rather than geometric uncertainty, it cannot serve directly as a depth confidence estimate for downstream decision-making. A robot relying on this uncertainty map to flag unreliable depth would correctly identify mirror surfaces as high-uncertainty, but incorrectly so from a geometric standpoint -- the depth error at those pixels may be large or small depending on whether the mirror hallucination aligns with the actual surface. This misalignment between photometric and geometric uncertainty is not a deficiency unique to this method; it is inherent to any uncertainty trained on photometric objectives.

Several limitations constrain the current results. The Booster training set contains 164 images across 27 scenes, which is small relative to the diversity of indoor reflective environments. The resolution sensitivity observed -- higher resolution performing worse -- suggests that the model is operating at the boundary of what this dataset size can support reliably. Training stability may also benefit from a warmup schedule or lower initial learning rate for the uncertainty branch. Finally, the three identified failure modes (mirrors, transparent surfaces, metallic highlights) each likely require different interventions beyond uncertainty-modulated training: mirror geometry prediction may benefit from explicit reflective surface labeling, and transparent surfaces may require transmission-aware photometric models.

7. CONCLUSION

This report presented an uncertainty-aware extension of Monodepth2 for self-supervised monocular depth estimation on indoor reflective surfaces. A Poggi-style per-pixel log-variance uncertainty head was added to the decoder and trained using a Laplacian NLL objective that automatically down-weights photometrically unreliable pixels. Fine-tuning on the Booster dataset showed consistent improvement across all seven standard depth metrics, with the best checkpoint achieving Abs Rel 0.255 -- a 7% relative improvement over the strongest fine-tuned baseline. Qualitative heatmaps confirmed that the uncertainty head learns to localize specular surfaces without any explicit annotation. Sparsification analysis established that photometric uncertainty and geometric uncertainty are distinct quantities, and that MC Dropout provides better-calibrated depth confidence at increased inference cost. The results suggest that uncertainty-modulated training is a practical and low-overhead approach to improving SS-MDE on non-Lambertian surfaces, and that pairing it with epistemic uncertainty estimation at inference time addresses the calibration limitation for applications requiring reliable depth confidence.

8. REFERENCES

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All experiments, implementation, training, and analysis were performed by Andrew Saifnoorian as an independent Deep Learning course project. Repository: github.com/andrewsaifnoorian | Branch: `uncertainty-masking-optimization`